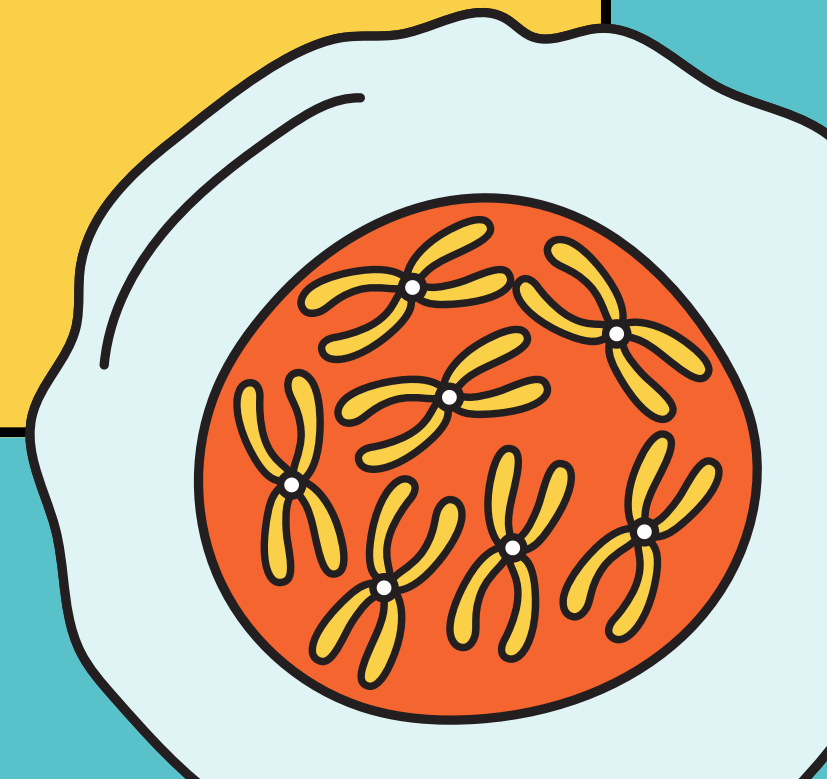




USES OF GENETICALLY MODIFIED ORGANISMS



GMO APPLICATIONS

- Genetic modification has extended its applications in the following fields:
 - agriculture;
 - animal production;
 - medicine; and
 - environmental protection.



AGRICULTURE

- GM crops have higher resistance to environmental stressors, can produce more nutrients, and have longer shelf life.



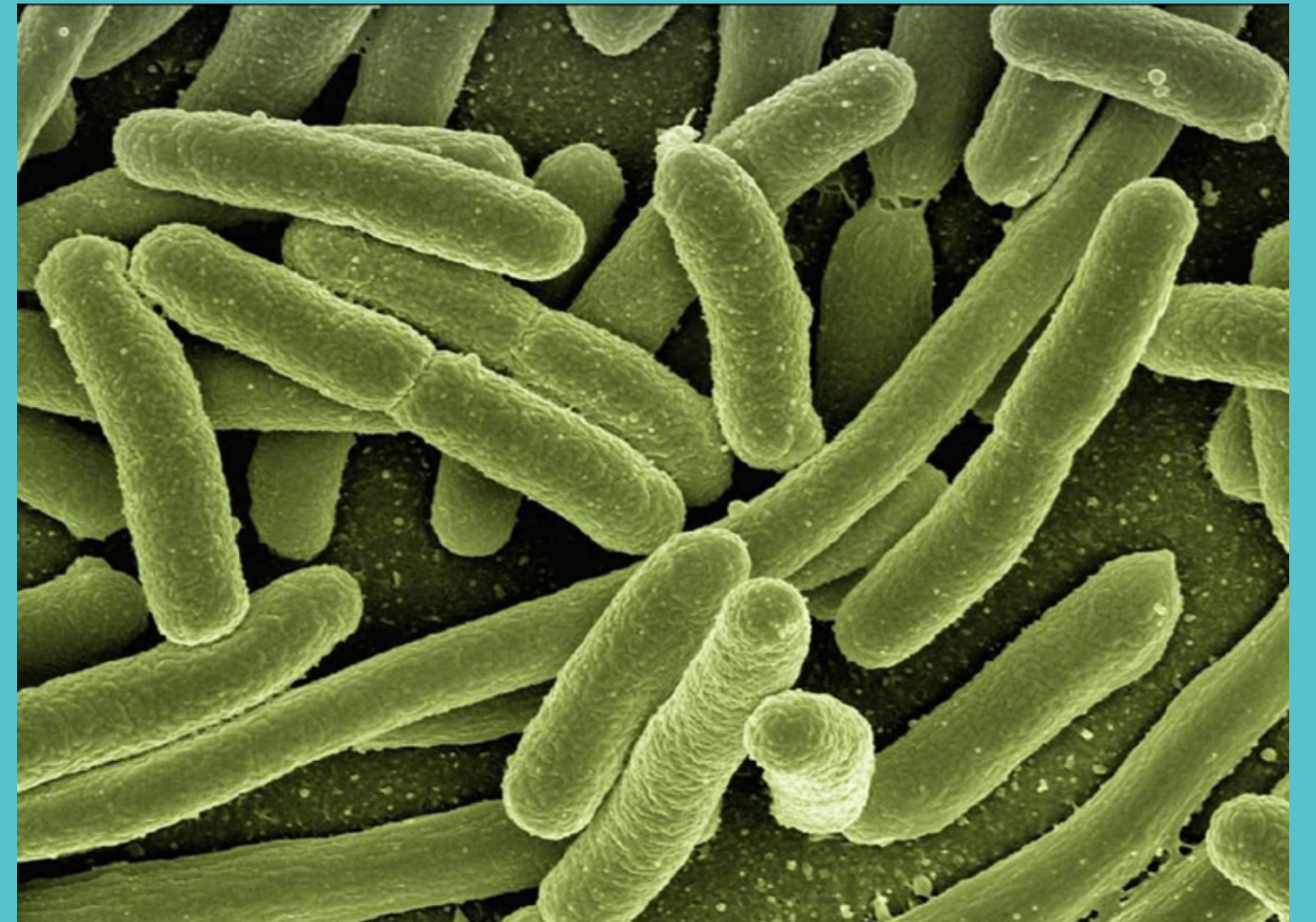
ANIMAL PRODUCTION

GM animals have improved meat quantity and quality and resistance to some diseases



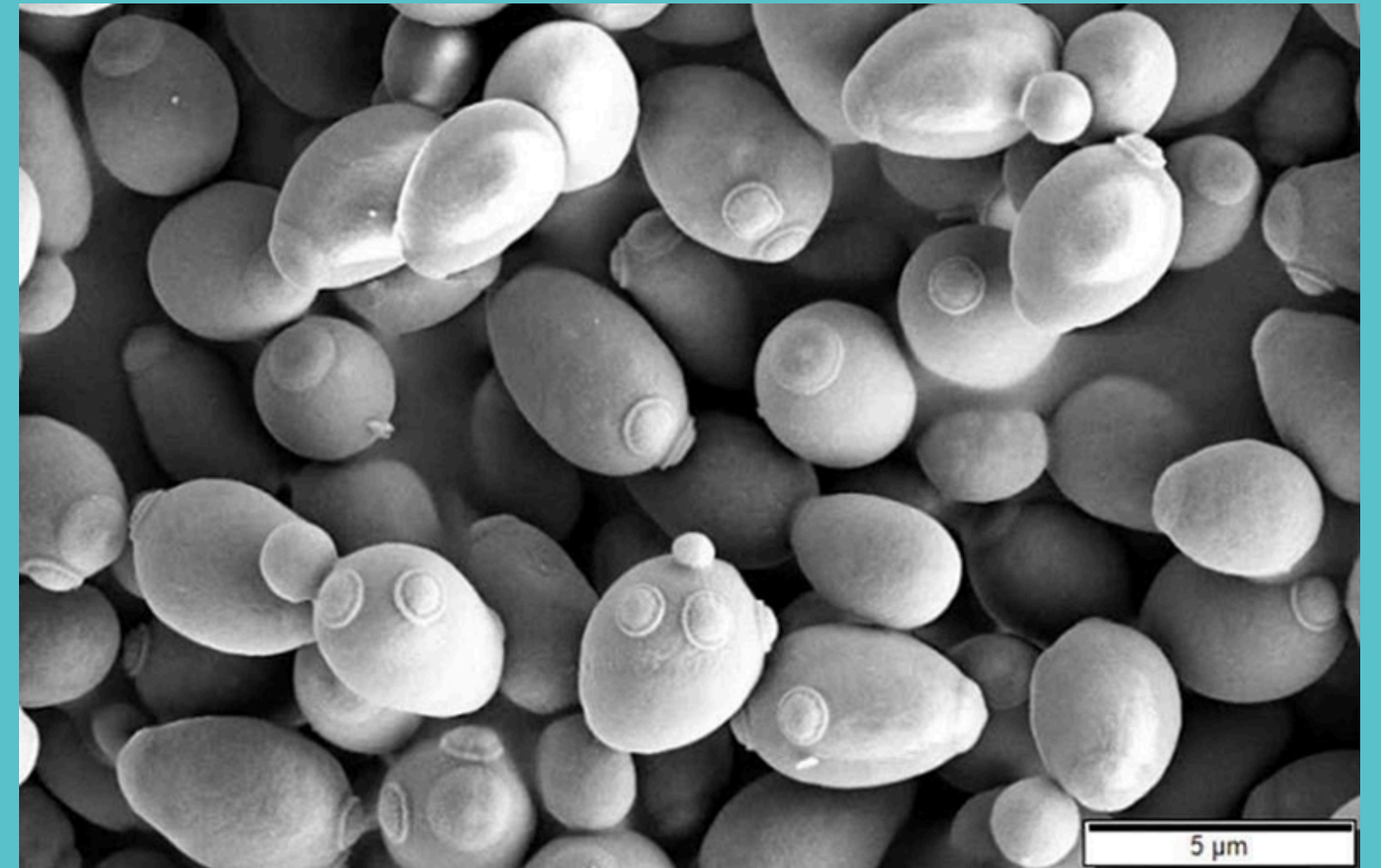
MEDICINE

Some GMOs can produce hormones, antibodies, and other therapeutic proteins for humans



ENVIRONMENTAL PROTECTION

Some GMOs have higher capacity
for bioremediation by processing
more environmental
contaminants



EXAMPLES OF GMO:

- Bt corn produces Bt toxin (cry1Ab protein) to deter Asian corn borer (*Ostrinia furnacalis*).
- Bt toxin gene comes from *Bacillus thuringiensis*.



EXAMPLES OF GMO:

- Golden Rice is modified to produce more beta carotene.
- Genes for phytoene synthase and carotene desaturase come from daffodils and bacteria, respectively.



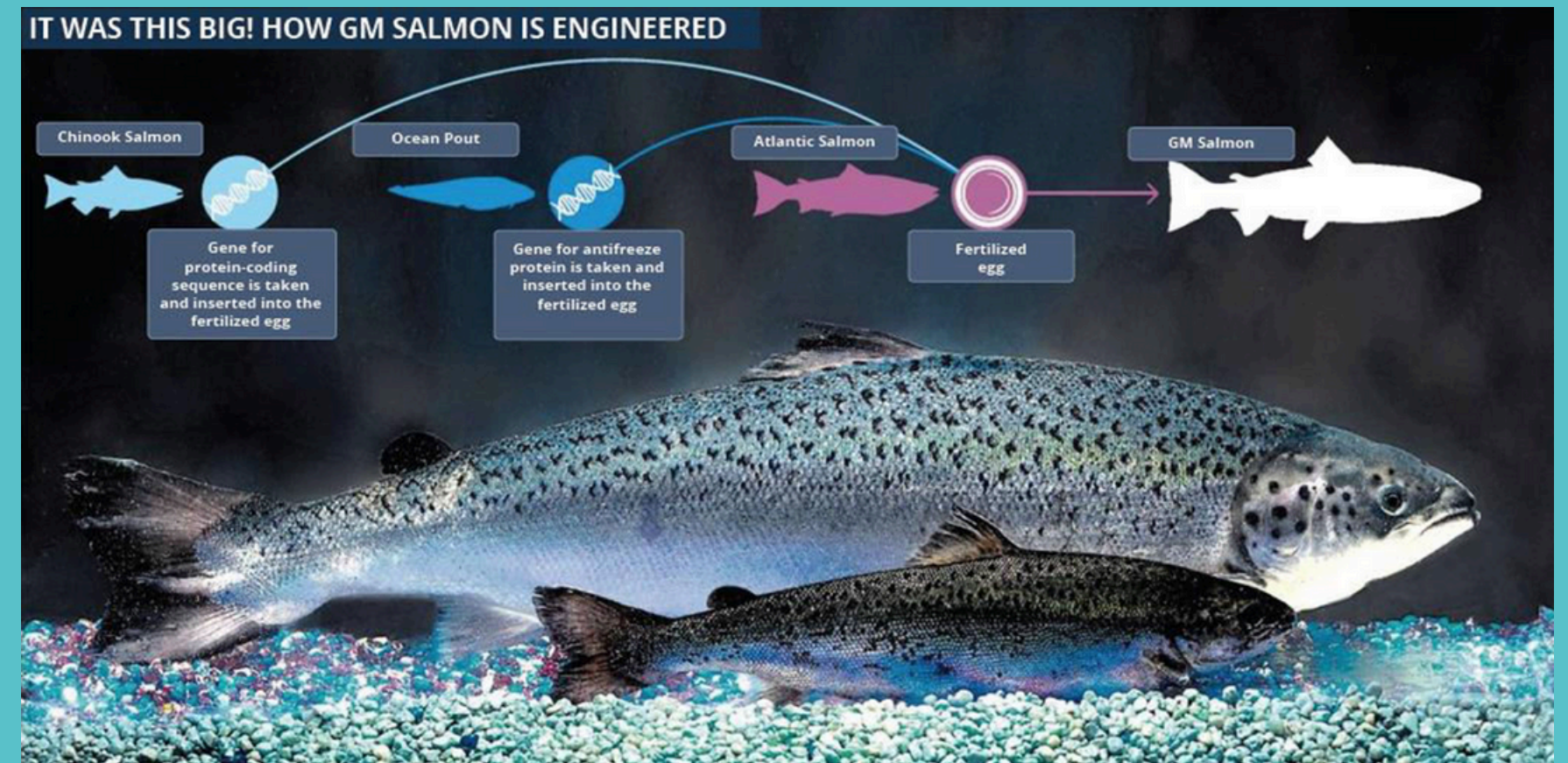
EXAMPLES OF GMO:

- FLAVR SAVR tomato
- Tomatoes are modified to have extended shelf life.
- It results from underexpression of polygalacturonase.



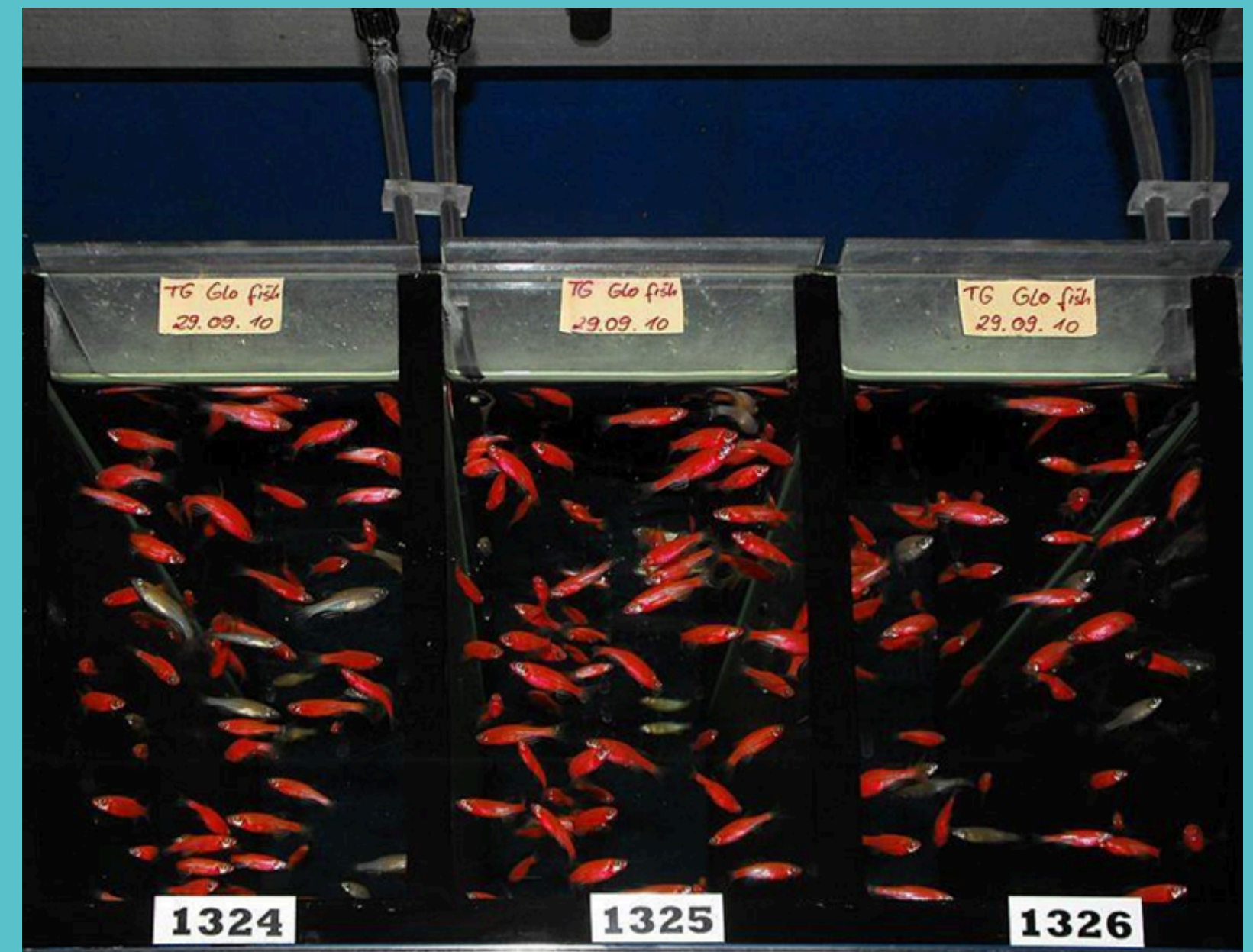
EXAMPLES OF GMO:

- Arctic salmon is modified to grow at higher rates. It has extra gene for growth hormone from the Chinook salmon



EXAMPLES OF GMO:

- GloFish
 - Danio rerio or zebra fish is modified to have fluorescent protein from sea anemone.
- It is a controversial product of genetic engineering



EXAMPLES OF GMO:

- Insulin-producing bacteria
- Insulin is now produced from transgenic bacteria to make it available for Type 1 diabetes patients.



EXAMPLES OF GMO:

- Bioreactor cows and sheep
- Cows and sheep are engineered to produce antibodies and therapeutic proteins for humans.



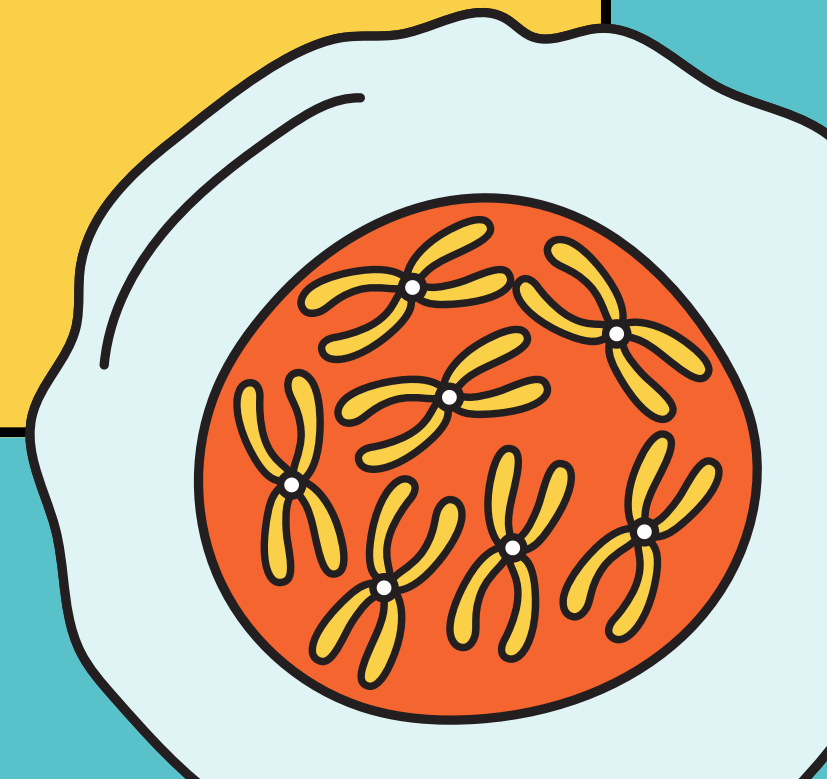
EXAMPLES OF GMO:

- Yeast and bacteria
- Microorganisms such as yeasts and bacteria are modified to have more resistance to extreme conditions.
- This enhances their capacity for bioremediation.





BENEFITS AND RISKS OF USING GMOS



BENEFITS – INCREASED PRODUCTIVITY

Genetic modification allows crops to have greater resistance to pest and diseases

- This enables farmers to have greater yield.



BENEFITS – INCREASED PRODUCTIVITY

Common GM crops in the Philippines include Bt corn and Bt talong.

- The Supreme Court is strictly regulating distribution of GM crops.



BENEFITS - IMPROVED NUTRITIONAL QUALITY

Genetic modification allows biosynthesis of more nutrients (usually vitamins A, C, D, and B complex) in GM crops



BENEFITS - IMPROVED NUTRITIONAL QUALITY

- This resolve food security issues in third-world countries.
- In the Philippines, improves crops help increase yields.



BENEFITS – DISEASE DIAGNOSIS

- Bioluminescent animals provide protein trackers for disease detection.
- This has been applied to different laboratory animals.



BENEFITS – DISEASE DIAGNOSIS

- They help in the detection of diseases that normally don't show symptoms.
- This is highly helpful in disease research (ex. cancer research).



RISKS - REDUCTION IN INSECT BIODIVERSITY

- Some non-target and non-damaging insects also become affected by pest-resistant plants.
- Bt pollen may affect monarch caterpillars.



RISKS – ALLERGIC REACTIONS

- Because of modifications in the proteins produced, some GMO products have caused allergic reactions in humans.



RISKS – EXPENSIVE SEEDS

- Farmers cannot use second generation seeds of GM crops.
- Farmers have to purchase GM crop seeds every year compliant to patent rights of the producing company



RISKS – RELIGION AND ETHICS

- Existence of GMOs is being questioned around the globe.
- In the Philippines, acceptance towards GMO is still on progress.



RISKS – RELIGION AND ETHICS

- Activists and experts state that scientists have no right to manipulate organisms.

